MILITARY SPECIFICATION

CANNULA SET, LACHRYMAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a set of three lachrymal cannulae suitable for introduction into the lachrymal duct.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-up.

Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards and drawing required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z70.1	American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Material. Cannulae and cleaning wires shall be fabricated from class 3, type 304 material in accordance with GG-I-526. The hubs and extension members shall be fabricated from class 8, type 260, 270 or 360 material in accordance with GG-I-526.

3.2 Hardness. The Rockwell hardness of the class 3 material, excluding cleaning wires, shall be B75 to B85.

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3.3 Style, design and dimensions. The cannula set shall consist of three lachrymal cannulae suitable for medical procedures involving the lachrymal duct. There shall be one reinforced straight cannula, one reinforced curved cannula, and one straight cannula. Each cannula shall be provided with an assembled cleaning wire. The hubs of the cannulae shall be the female Luer type in accordance with ANSI Z70.1 and shall meet the requirements of 3.6. The cannulae hubs shall be capable of engaging with both Luer Lock syringes and Luer Slip syringes. The hubs shall be shaped not to roll and to provide a satisfactory finger grip for attachment to syringe. The dimensions of the cannulae shall be in accordance with Figure 1.

3.4 Finish. Finish shall be bright and smooth.

3.5 Passivation. All components fabricated from class 3 material shall be properly passivated.

3.6 Diameter and depth of Luer hub cavities. The cavities of the female Luer Lock hubs shall be tested for diameter and depth of the Luer cavity as specified in 4.4.1. Cavities which do not fit the plug gage within the "GO-NO GO" limits specified in 4.4.1 shall be rejected. In addition, when the hub cavity is fitted onto the plug gage, there shall be no movement of the cavity on the gage when the hub is pressed lightly from side to side in all directions.

3.7 Fluid tightness. When tested as specified in 4.4.2, the female Luer Lock hubs, and all junctions of the cannula proper shall not show leakage exceeding the formation of one clinging drop.

3.8 Security of assembly. When tested as specified in 4.4.3, the cannula shall not separate or loosen from the hub or extension, or the extension from the hub.

3.9 Freedom from foreign matter. All hub cavities shall be tested for freedom from foreign matter, as specified in 4.4.4, and shall show complete freedom from metal chips and other foreign matter.

3.10 Identification marking. Cannulae shall be marked with the name or registered trademark of the contractor. Cannulae of other than U.S. manufacturers shall also be marked with the country of origin.

3.11 Workmanship. Workmanship shall be first class throughout. Cannulae shall be free from burrs, sharp edges, cracks, holes, pits, scale, corrosion, flaking or incomplete plating, foreign particles, and any other defects which detract from their appearance or impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the corrosion-resisting steel alloy and copper alloys may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table I. The unit of product for examination purposes shall be one cannula set.

Table I. Sampling Criteria For Examination

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. The unit of product for test purposes shall be one cannula set. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Table II. Sampling Criteria for Tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
Cannula Set	Boil test (all material)	3.1 & GG-I-526	GG-I-526	S-2	1.5
	Salt-spray test	3.1 & GG-I-526	GG-I-526	S-2	1.5
	Plating test	3.1 & GG-I-526	GG-I-526	S-2	1.5
	Hardness	3.2	GG-I-526	S-2	2.5
	Dimensions	3.3	----	S-2	2.5
	Diameter and depth of Luer hub cavity	3.6	4.4.1	S-4	1.0
	Fluid tightness test	3.7	4.4.2	S-4	1.0
	Security of assembly	3.8	4.4.3	S-2	1.5
	Freedom from foreign matter	3.9	4.4.4	S-2	1.5

4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in the construction of the cannula set shall be selected for chemical analysis.

4.3 Examination. The cannulae shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any cannula in the sample containing one or more defects shall be rejected and if the number of defective cannulae in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Cannulae not of design specified.
102	Cannulae set not furnished complete with all specified components.
103	Swaged and soldered joints not free from cracks, crevices, fissures, flux, discoloration and excess solder.
104	Cannulae not free from sharp edges, burrs, dents, cracks, pits, rough areas and tool marks.
105	Finish not bright and smooth.
106	Plated parts not free from flaking and not uniform (including hub cavities).

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Table III. Classification of Defects (cont'd)

Categories	Defects
Minor	
201	Cannulae not identified as specified in 3.10.
202	Cleaning wire not assembled to each cannula.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Diameter and depth of Luer hub cavities. The Luer hub cavities of the cannulae shall be tested with Gages for Female Luer Gages for Female Luer Connections in accordance with ANSI Z70.1. This gage shall be as shown in the ANSI Standard, except that it shall be constructed with a suitable fluid path to permit performance of the fluid tightness test specified in 4.4.2. The Luer hub cavity shall be placed onto this gage with a straight motion, without turning or twisting the plug gage into the cavity. Once the Luer hub cavity is placed onto the gage, no additional force shall be exerted to force the plug gage into the Luer cavity. The fit of the Luer hub cavity onto this gage shall be tested by means of the feeler gage in accordance ANSI Z70.1.

4.4.2 Fluid tightness test. Samples of the cannulae shall be subjected to an internal water pressure of 45 pounds per square inch gage (psig). The plug gage specified in 4.4.1 shall be inserted into the female Luer hub cavity with a straight motion, without turning or twisting the plug gage into the cavity with a force not to exceed 5 pounds. The feeler gage shall be placed onto the plug gage to assure that the "GO" position is not violated. The cannula shall be placed in a horizontal position with the cannula lumen sealed at the distal end in a leakproof manner. Water shall then be internally admitted to the testing arrangement in a leakproof manner until the entire cannula lumen and hub cavity is filled (air bubbles eliminated) with internal pressure of 45 psig for 30 seconds. The entire assembly shall be noted for leakage as specified in 3.7.

4.4.3 Security of assembly. The cannulae shall be tested for security of assembly by applying a straight force, or pull, of 25 pounds to the cannula or extension for a period of at least 1 minute with the hub appropriately secured.

4.4.4 Freedom from foreign matter. The Luer hub cavities shall be tested for freedom from foreign matter by visual examination utilizing a magnifier and good illumination. A cotton swab, with cotton fibers tightly wrapped around a suitable applicator, shall be vigorously turned down into and around the internal surfaces of the hub cavity. The cotton fibers shall be examined for metal chips and other foreign matter utilizing a magnifier.

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4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV. Examination of Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One set containing three lachrymal cannulae, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each cannula shall be thoroughly cleaned, wrapped in neutral tissue or other suitable non-corrosive material and the complete set packaged in accordance with MIL-P-116, method IC-2 except that a preservative shall not be utilized.

5.1.2.2 Intermediate package. Twelve unit packages shall be packaged in an intermediate package constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure of the intermediate container shall be adequate to prevent accidental opening.

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5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Thirty-six intermediate packages (432 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable box specification.

5.2.2 Level B. Thirty-six intermediate packages (432 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method I or II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lachrymal cannulae are intended for introduction into the lachrymal duct.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this document.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Marking requirements (see 5.3).

6.3 This specification is intended to cover the following items listed in the Federal Supply Catalog:

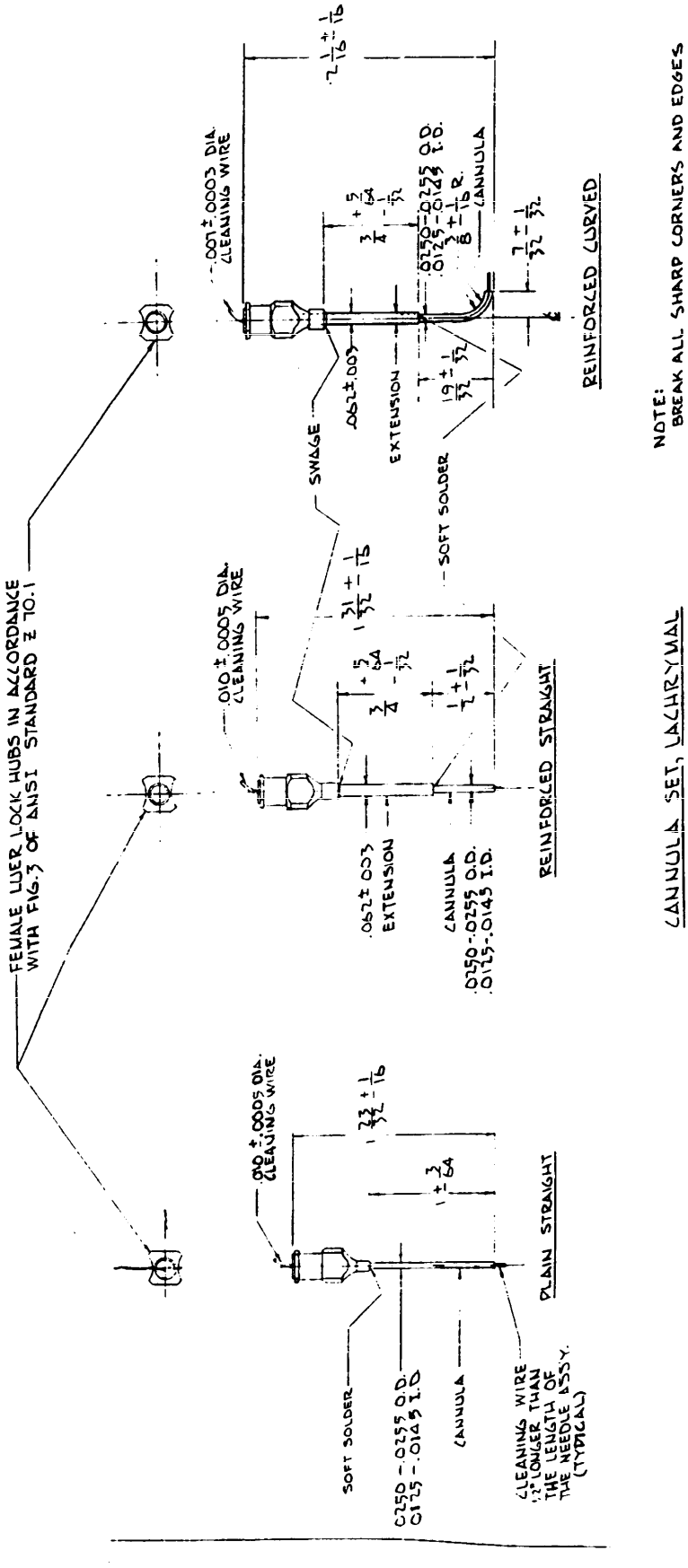
NSN	Item Identification
6515-00-314-0400	CANNULA SET, LACHRYMAL

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity
Defense Supply Agency - DM

Project No. 6515-1905
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CANNULA SET, LACHRYMAL

FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-37127 CANNULA SET, LACHRYMAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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